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Machine Learning-based Scientific Article Similarity Checking Research

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Abbreviations

VSM – Vector Space Model

SVM – Support Vector Machine

CNN – Convolutional Neural Network

RNN – Recurrent Neural Network

LCS – Longest Common Subsequence

PAN – Plagiarism Analysis

NLP – Natural Language Processing

LSTM – Long Short-Term Memory

BERT – Bidirectional Encoder Representations from Transformers

SBERT – Sentence-BERT

TF-IDF – Term Frequency-Inverse Document Frequency

CL-PD – Cross-Lingual Plagiarism Detection

SimCSE – Simple Contrastive Learning of Science Embeddings

1. INTRODUCTION

Scientific inquiry has changed fundamentally. This shift, primarily fueled by the modernization of academic libraries and the sudden explosion of open-access repositories, serves as a double-edged sword. While it certainly facilitates a vital global exchange of knowledge, it has unintentionally lowered the barrier for academic misconduct, most notably plagiarism. As the volume of digital content continues to grow, the strategies used to mask stolen work have become incredibly sophisticated. Plagiarists no longer rely on simple "copy-paste" methods that traditional tools flag instantly. Instead, they hide the original source through a complex web of paraphrasing, synonym replacement, and structural reordering. Consequently, unmasking this "semantic plagiarism" has become a defining challenge for today's academic community (Malik et al., 2021).

Traditional plagiarism detection systems Standard detection systems frequently fall short. The reason is simple: they rely too heavily on lexical string matching and fingerprinting algorithms. While these systems are undeniably effective at spotting exact matches, they often fail to grasp the deeper, underlying intent of a text (El-Rashidy et al., 2022). To address this limitation, recent research is pivoting toward the intersection of Artificial Intelligence and Natural Language Processing (NLP). By leveraging techniques like Deep Learning and Transformer-based models, we can now analyze text at a semantic level, capturing context rather than just surface-level syntax (Liu et al., 2023). However, applying these models to scientific articles is not easy. The sheer length, technical complexity, and specific formatting of academic texts present unique computational hurdles. This master thesis addresses these specific obstacles by developing a Machine Learning-based approach tailored specifically for scientific article similarity checking.

Recently, the problem has expanded beyond human-driven deception to include machine-generated content. With the rapid advancement of Large Language Models (LLMs), scientific publishing now faces a genuine threat from synthetic text that mimics academic styles but lacks any foundation in authentic research. (Alhijawi et al., 2025) identify this as a critical modern challenge. They observe that standard similarity tools often fail to distinguish human manuscripts from those produced by models like ChatGPT, which can generate coherent but entirely fabricated data. This shift demands a total evolution in our detection strategies. As (Greiner-Petter et al., 2025) argue in their analysis of the PAN 2025 shared task, robust deep learning models are no longer a luxury; they are essential for spotting the sophisticated generative plagiarism that slips past naive semantic checks.

1.1 Investigation Object

The Investigation Object is machine learning and natural language processing algorithms designed for semantic text similarity assessment and plagiarism detection within scientific literature.

1.2 The Aim and Tasks of the Thesis

The Aim of the Research is to improve the detection of semantic plagiarism in scientific articles by developing a hybrid similarity checking model that integrates deep learning embeddings with structural feature analysis.

Three (3) tasks were defined to achieve the aim of the thesis. They are as follows:

1. To analyze and compare state-of-the-art text similarity detection methods, specifically evaluating the performance of Deep Learning architectures (such as BERT and LSTMs) against Feature Engineering techniques (such as SVM).
2. To propose a scientific article similarity approach that combines semantic vector representations (e.g., SBERT) with syntactic alignment algorithms to enhance detection accuracy for paraphrased content.
3. To implement the proposed approach and experimentally evaluate its validity and performance metrics against existing baseline models.

1.3 Novelty of the Topic

The novelty of this research is grounded in addressing the detection gaps that currently exist between machine-generated text and complex human obfuscation. Traditional systems do a decent job at catching verbatim copying, yet (Greiner-Petter et al., 2025) point out in their PAN 2025 overview that most semantic models still stumble when trying to generalize against content from LLMs like Llama or Mistral. It is a growing problem. Furthermore, (Alhijawi et al., 2025) even suggest that we now need entirely distinct deep learning architectures just to tell the difference between real scientific inquiry and the "hallucinated" data ChatGPT spits out. Most current checkers simply lack this capability. Furthermore, because existing algorithms usually assume a simple sentence-to-sentence alignment, they miss the bigger picture. However, (Saqaabi et al., 2025) proved this by showing that sentence-level detectors fail during paragraph-level manipulation where text is reordered or split across a wider context window. By moving our analysis from isolated lines to contextual

paragraphs, this thesis introduces a hybrid model designed to capture those hidden inter-sentence dependencies that currently slip past state-of-the-art systems.

1.4 Relevance of the Topic

The relevance of this research is driven by the massive pressure on academic institutions and journals to maintain the integrity of the scientific record.

It is no longer a minor concern. (Alnaimat et al., 2025) highlight that academic journals are currently drowning in a sea of submissions that require intense ethics checks. Automated tools are the only way to keep scientific credibility intact. This necessity is statistically supported by (Murch et al., 2025), who analyzed recent doctoral dissertations and found that roughly 6% of the text was AI-generated, which is a clear sign that misconduct is changing. Because of this, (Setu et al., 2025) contend that combining semantic checks with direct copy detection is now a requirement for fair assessment. Their findings validate why developing the robust detection model in this work is so practically important.

1.5 Research Methodology

The analytical side of this work relies on information analysis and comparative research to map out the current state of detection techniques, focusing specifically on the trade-offs between feature-based classifiers like SVM and neural networks such as CNN or LSTM.

To build the proposed approach, logical generalization and conceptual modelling were used to design a system architecture that really improves the gaps in semantic analysis. For the practical side, experimental research methods were used by testing the prototype against benchmark datasets like PAN 2011 and 2013 to get a clear, quantitative measure of how much improvements were made in Precision, Recall, and F1-scores.

1.6 Scientific Value of the Thesis

The main scientific value of this work lies in the development of an optimized workflow for scientific text similarity checking. By rigorously comparing and integrating Deep Learning embeddings with feature-based heuristics, this thesis contributes new knowledge regarding the effective application of NLP in the domain of academic integrity. The research provides a validated methodology for detecting non-verbatim plagiarism, offering a more robust alternative to existing string-matching paradigms.

1.7 Main Results of the Thesis

A systematic comparison of six contemporary machine learning approaches for plagiarism detection, establishing the superior efficacy of Transformer-based models for semantic analysis over traditional lexical methods.

1.8 Structure of the Work

The first section introduces the context of scientific plagiarism and defines the research problem and objectives. The second section deals with the analysis of related works, reviewing methodological approaches from recent literature and standard evaluation datasets. The third section will describe the design of the proposed similarity detection architecture and the specific algorithms selected for implementation. The fourth chapter will describe the experimental validation of the developed prototype and discusses the obtained results.

1.9 Approbation of the work

This section describes the planned approbation of the work, i.e., how the results of the thesis will be presented and disseminated.

The approbation of this research will be carried out after the main results of the thesis are obtained. It is planned to prepare a scientific presentation and participate in a conference related to information systems or artificial intelligence, such as the annual “Information Systems and Technologies” conference organized by Vilnius Gediminas Technical University:
https://conferences.vilniustech.lt/index.php/SITIS/en

2. RELATED WORK ANALYSIS

The surge in digital academic resources and the growth of open-access repositories has completely reshaped the world of scientific research. Even though this accessibility promotes a vital global sharing of knowledge, it has also, quite unintentionally, led to a rise in academic misconduct, especially plagiarism. This part of the thesis analyzes current scientific literature to trace how detection methods have changed over time. We will look at the shift from basic feature engineering to the much more advanced Deep Learning architectures used today.

2.1 Description of Plagiarism Detection and Taxonomy

At the very core, plagiarism is defined as the act of taking someone else's words, ideas, or work and passing them off as your own without giving proper credit (Smith, 2023). This breach of academic and professional integrity goes far beyond simple copying and pasting. It is important to remember that plagiarism isn't always a deliberate act. Often, accidental citation omissions or disorganized research notes can lead to serious plagiarism charges. This makes the rigorous acknowledgment of every source vital in every piece of writing. Here is a clear example showing how to take an original sentence and rewrite it properly to avoid these issues:

Table 1. Example of Plagiarism and Paraprhasing (Smith, 2023)

Original Text	Plagiarism	Paraphrased
Imagine this sentence appeared in a book by an author named J. Smith (2023): "The swift expansion of artificial intelligence in the workplace has created a demand for new skills, causing many traditional roles to become obsolete."	"The fast growth of AI in offices has made a need for fresh skills, making many old jobs outdated." **Why this is plagiarism.** This is called "patchwriting." Even though the writer changed a few words ("swift" to "fast", "workplace" to "offices"), the sentence structure remains the same as the original. It also lacks a citation.	"According to (Smith, 2023), the job market is shifting as automation replaces older professions, requiring employees to learn entirely different capabilities." **Why this works.** 1. Structure Change: The sentence structure is completely different from the original. 2. New Vocabulary: It conveys the same meaning but uses unique words ("job market is shifting," "automation," "capabilities"). 3. Citation: It explicitly gives credit to the original source (Smith, 2023).

To understand the current scope of plagiarism is the first step toward understanding why detection is so difficult today (Amirzhanov et al., 2025), note in their systematic survey that plagiarism involves the unacknowledged use of another person's language, ideas, or results. In our current academic environment, the problem has shifted. It is no longer just about simple copy-pasting but has moved into much more complex types of obfuscation. Current research usually sorts these into the two main categories shown in Figure 1:

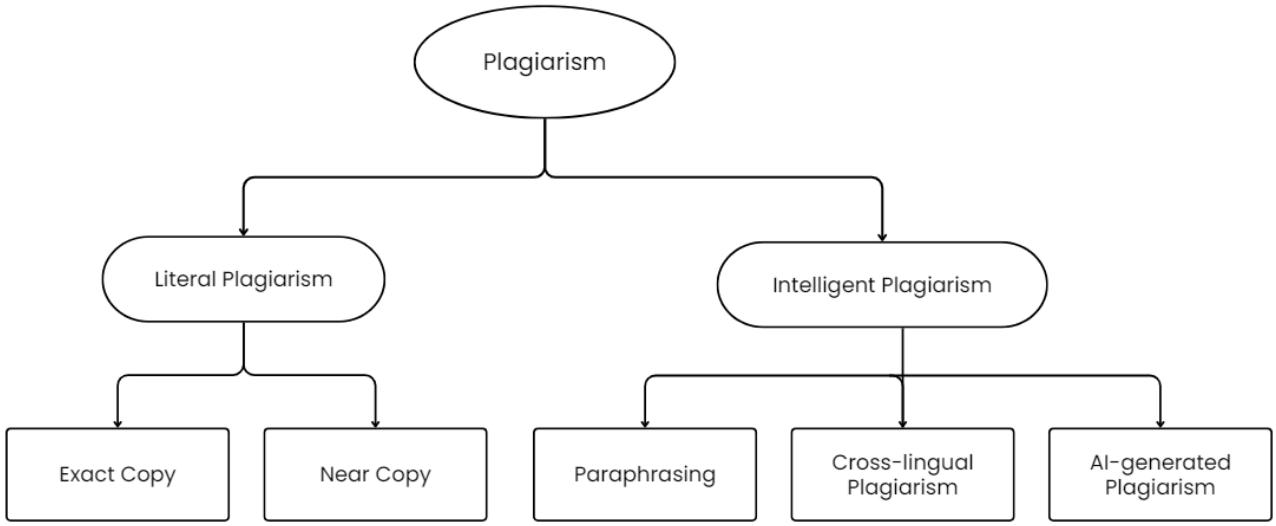

Figure 1. Plagiarism Classification from (Amirzhanov et al., 2025; Lieharyani et al., 2025)

1. Literal Plagiarism.

Literal plagiarism is the most basic form of academic fraud. Often called verbatim or copy paste plagiarism, it is the fundamental type of misconduct researchers face today. (Amirzhanov et al., 2025) recently described this in a systematic taxonomy as the direct use of text segments from a source without enough changes or credit. It might be the simplest way to hide stolen work, but it stays common because automated text scraping is so easy (Setu et al., 2025) make an important point that this acts as a first layer for detection. Algorithms have to find these direct overlaps first. Only then do they move on to the more expensive semantic models.

Literal plagiarism is further categorized into two distinct sub-types:

1. **Exact Copy (Verbatim).** This is exactly what it sounds like. It involves taking a sentence or an entire document and copying it word for word. Every single character and the entire sequence of text stays identical to the original source.
2. **Near-Copy (Camouflaged).** In this case, the plagiarist tries to trick simple string matching filters by making small, surface level changes. (Lieharyani et al., 2025) call these tweaks lexical

noise. This might include removing commas, fixing spelling, or changing "do not" to "don't." Even with these edits, the actual lexical structure is statistically the same as the original.

2. Intelligent (Obfuscated) Plagiarism. Intelligent plagiarism is a much more sophisticated beast. Often called obfuscated or semantic plagiarism, it is a clever attempt to hide where content came from by messing with the linguistic structure. The key here is that the core meaning stays the same. Unlike the literal kind, this does not rely on simple word overlaps. Instead, it breaks the sequence of words. This makes traditional string-matching algorithms almost ineffective. (Amirzhanov et al., 2025) classify this as a semantic challenge. The goal for the plagiarist is to change the fingerprint of the text without actually changing the message. (Sajid et al., 2025) note that detecting this requires very heavy Natural Language Processing (NLP) models like Transformers. These tools have to capture the underlying semantic vector space rather than just looking for surface level keywords.

Based on the analyzed literatures below, intelligent plagiarism is categorized into three distinct classifications:

1. **Paraphrasing.** Paraphrasing is more than just swapping words. It is a deliberate effort to rewrite text using entirely different sentence structures and synonyms while making sure the original meaning stays intact. This creates a massive headache for detection. (Saqaabi et al., 2025) highlight that complex paragraph-level manipulations, such as splitting, joining, or shifting sentences, render standard sentence-matching models ineffective. This method involves translating a text from a source language into a target language and claiming it as original work.
    i. **Mechanism.** Instead of merely swapping synonyms in a single sentence, plagiarists now employ structural strategies such as splitting one sentence into two, joining multiple sentences, or shifting the order of sentences within a paragraph.
    ii. **Detection Challenge.** (Zhou et al., 2024) highlight that standard datasets often lack these complex samples, causing deep learning models to fail when the syntactic structure is heavily scrambled.
2. **Cross-Lingual Plagiarism.** Translating a text from one language (e.g., Urdu, Bengali, Marathi) into another to evade monolingual detectors (Bouaine et al., 2023; Zahid et al., 2023). is type involves translating a text from a source language into a target language and claiming it as original work.
    i. **Mechanism.** A plagiarist might translate a scientific paper from English to Spanish, or Urdu to English to mask the origin.

ii. **Detection Challenge.** (Bouaine et al., 2023) and (Zahid et al., 2023) emphasize that this method completely removes lexical overlap. Detection requires "multilingual embedding spaces" where the vector for the English word "Water" is mathematically close to the Spanish word "Agua," or the use of intermediate translation layers (Islam et al., 2023).

3. **AI-Generated Plagiarism.** The emergence of Large Language Models (LLMs) has introduced a new, highly elusive category.
   i. **Mechanism.** Users prompt models like ChatGPT or Llama to "rewrite this paragraph in a scientific style" or "generate a literature review on topic X." (Alhijawi et al., 2025) describe this as the generation of "synthetic scientific content" that mimics the statistical distribution of human writing but lacks authentic provenance.
   ii. **Detection Challenge.** (Greiner-Petter et al., 2025) warn that this content is unique at the character level (passing literal checks) and semantically coherent (passing simple semantic checks), requiring distinct deep learning classifiers to identify "hallucinated" patterns.

2.2 Relevance to Scientific Article Similarity Checking

Detecting plagiarism in scientific papers is a different beast entirely compared to standard text analysis. The core issue lies in the way academic writing is actually built. These papers follow very rigid structures and use a high density of specialized terms. This leads to a specific context similarity problem where basic lexical detectors might flag standard technical jargon as actual collusion (Lieharyani et al., 2025). To protect the integrity of academic work, we need broad strategies. Simple string matching is just not enough to tell the difference between a field's standard vocabulary and actual content theft (Setu et al., 2025).

Additionally, Complex arguments also make scientific papers vulnerable to smart obfuscation. For example, authors might use paragraph level paraphrasing like splitting or shifting sentences. Such moves make traditional sentence level similarity checks ineffective without the help of deep learning (Saqaabi et al., 2025). The rise of generative AI makes this situation even more difficult. Today, Large Language Models can synthesize scientific content that looks plausible. This forces us to find new detection paradigms to verify that doctoral dissertations and research papers are authentic (Alhijawi et al., 2025; Murch et al., 2025). Thus, modern similarity checking has to evolve. We need robust semantic models that can navigate the complicated linguistic world of scientific literature (Greiner-Petter et al., 2025).

2.2.1 Comparison of Detection Methods in Analyzed Literature

Table 2 Comparisn of Detection Methodscategorizes traditional scientific article similarity checking methods into lexical matching and feature-based approaches, while new methods leverage deep learning, Transformers, and optimization techniques. Traditional methods excel in efficiency and interpretability, making them suitable for detecting verbatim copying, but they are limited by their inability to understand semantic context. Conversely, new methods offer superior accuracy and robustness against intelligent plagiarism like paraphrasing, though they come with higher computational costs and complexity.

Table 2 Comparisn of Detection Methods

Traditional Methods				
Method Type	**Key Approaches**	**Strengths**	**Limitations**	**Citations**
Lexical & Statistical	N-gram Overlap: Checks direct word sequences.	Very fast computation, suitable for large databases.	Fails to detect paraphrasing or synonym replacement.	Lieharyani et al. (2025)
	Vector Space Model (VSM): Represents text as vectors based on word frequency.	Easy to understand why a match was flagged (e.g., "these 5 words are identical").	Struggles in homogenous datasets (e.g., scientific papers with shared jargon), leading to false positives.	Arabi & Akbari (2022)
	Levenshtein Distance: Measures edit distance between strings.	Effective for "Literal Plagiarism" (copy-paste).		Sajid et al. (2025)
Feature-Based Machine Learning	SVM Classifiers: Uses engineered features (sentence length, POS tags) to classify text.	Can detect stylistic anomalies better than simple string matching.	Requires domain expertise to define features.	El-Rashidy et al. (2024)
	Naïve Bayes: Probabilistic classification based on feature occurrence.	Feature importance can be analyzed (e.g., "too many adjectives").	Still relies largely on surface-level patterns rather than deep meaning.	Zahid et al. (2023)
New Methods				
Method Type	**Key Approaches**	**Strengths**	**Limitations**	**Citations**
Sequential Deep Learning	Siamese LSTM/RNN: Processes text sequences to learn context.	Captures sequential dependencies (word order matters).	Slower training than SVM.	Bao et al. (2024)
	Attention Mechanisms: Focuses on "salient" words in a sentence.	Better than ML at identifying reordered sentences.	Can struggle with very long scientific paragraphs without attention.	Asalkar et al. (2025); Bouaine et al. (2023)
Transformer-Based (SOTA)	BERT / E-BERT: Bidirectional semantic understanding.	Understands meaning, synonyms, and context (State-of-the-Art accuracy).	Requires GPUs and significant memory.	Antonius et al. (2023)

	Hybrid Models: BERT + Alignment (Smith-Waterman).	Highly effective against "Intelligent Plagiarism" and cross-lingual copying.	"Black box" nature makes it harder to explain specific matches to users.	Moravvej et al. (2023)
	Cross-Lingual Models: Multilingual BERT / Doc2Vec.			Rosu et al. (2020)
Optimization & Efficiency	DistilBERT: Smaller, faster Transformer.	Retains high accuracy while reducing computational time (critical for checking long articles).	Slightly lower accuracy than full BERT models (e.g., 97% vs 99%).	Jing & Liu (2023)
	Scenario-Based Selection: Switching between models (SimCSE vs SBERT) based on complexity.			Liu et al. (2023)

As can be summarized from Table 2, the categorization of detection methodologies reveals a fundamental trade-off between computational efficiency and semantic depth. Traditional Methods, relying on lexical matching (e.g., Levenshtein distance) and feature-based Machine Learning (e.g., SVM), remain highly effective for detecting verbatim copying and stylistic anomalies. (El-Rashidy et al., 2024; Lieharyani et al., 2025) demonstrate that these approaches offer superior speed and interpretability which makes them suitable for large-scale initial filtering, even though they theoretically fail against complex semantic obfuscation. In contrast, New Methods leverage Deep Learning and Transformer architectures (e.g., BERT, LSTM) to catch the underlying meaning of text. (Antonius et al.,2023.; Saqaabi et al., 2025) validate that these models provide state-of-the-art accuracy for detecting "intelligent" plagiarism, such as paraphrasing and cross-lingual translation; by analyzing bidirectional context, albeit at a significantly higher computational cost that recent optimization studies (Jing & Liu, 2023) are striving to mitigate.

2.2.2 Evolution of Detected Methods in Analyzed Literature

The field of plagiarism detection is currently witnessing a total paradigm shift. We are seeing a move away from the old days of simple lexical matching toward a world of incredibly complex Machine Learning (ML) and Deep Learning (DL) architectures. This change is not happening by accident. It is driven by a real need to catch "intelligent" plagiarism. This includes things like clever paraphrasing, hidden translations, and generative AI content that older, more traditional algorithms simply cannot find. Based on recent literature, this progression is happening in four distinct methodological stages:

1. Feature-Based Machine Learning. While Deep Learning dominates current research, traditional ML algorithms are still very relevant for their interpretability and efficiency. Authors of (El-Rashidy et al., 2024) demonstrated that when Support Vector Machines (SVM) is augmented with aggressive feature selection

(using Chi-square to select 42 stylistic features like POS tag ratios and sentence length), it can achieve a robust accuracy of 96.5%. This proves that "shallow" learning is still effective to identify stylistic anomalies in text if the features are carefully engineered. Similarly, (Zahid et al., 2023) successfully applied supervised classifiers (SVM, Naïve Bayes) to the challenging domain of Cross-Lingual Plagiarism Detection (CL-PD) and identified Urdu-English plagiarism by analyzing translated feature sets.

2. Sequential Deep Learning (RNNs and LSTMs). To overcome the limitations of manual feature engineering, researchers turned to Recurrent Neural Networks (RNNs). These models can learn patterns directly from raw text sequences. (Bao et al., 2024) introduced an Attentive Siamese LSTM. This allows the model to focus on specific "salient" words within a sentence pair, which significantly improving detection in low-resource languages. (Asalkar et al., 2025) provided a comparative analysis confirming that these LSTM-based models outperform Convolutional Neural Networks (CNNs) when it comes to capturing the long-range sequential dependencies typical of scientific arguments. (Bouaine et al., 2023) combined Doc2Vec embeddings with Siamese LSTMs and established a new benchmark for cross-lingual detection (English - Spanish) that surpasses standard BERT implementations in specific multilingual contexts.

3. Transformer-Based Models (The State-of-the-Art). The biggest advancement in this field is the adoption of Transformer architectures. They utilize self-attention mechanisms to understand the deep semantic context of text. (Antonius et al., 2023.) achieved a remarkable 99.5% accuracy by combining E-BERT (Enhanced BERT) with the classic Smith-Waterman alignment algorithm. This hybrid approach uses BERT's semantic understanding to "match" paraphrased concepts while using Smith-Waterman to precisely align the scrambled sentence structures a critical requirement for scientific text. (Moravvej et al., 2023) advanced this further by integrating Differential Evolution (DE) to optimize the weights of BERT and Attention-LSTMs, which effectively addresses the problem of "imbalanced datasets" where actual plagiarism cases are rare compared to original content.

4. Optimization and Efficiency. Full Transformer models are great, but they are also computationally expensive to process long scientific articles. That is the reason recent research has focused on efficiency. (Jing & Liu, 2023) proposed a population-based detection method using DistilBERT. It is a “distilled” version of BERT that keeps 97% of the performance but reduces training time by 60%. Similarly, (Liu et al., 2023) also developed a "scenario-based" framework that can choose between SimCSE (Simple Contrastive Learning of Sentence Embeddings) and SBERT depending on the complexity of the text. This ensures an optimal balance between detection speed and semantic accuracy.

In conclusion, while traditional ML still provides a high-speed baseline, the field is clearly moving towards Hybrid Transformer Models. By combining semantic strength of BERT with the structural precision of alignment algorithms, recent research is now able to detect the most subtle forms of scientific plagiarism.

2.3 Comparative Analysis of Related Literature Works

Here we present a comprehensive analysis of scientific articles related to scientific article similarity checking in tabular form. This comparative analysis details the specific machine learning or deep learning approach utilized in each study, the distinct problem domain addressed, ranging from standard text changes to cross-lingual and AI-generated plagiarism, and the standard benchmark datasets employed for validation. Furthermore, it identifies the key attributes used for prediction and summarizes the main experimental results, providing a clear overview of the current state-of-the-art in scientific article similarity checking.

Table 3. Comparative Analysis of Related Literature Work

Reference	Used Approach	Problem Domain Studied	Dataset Used	Attributes Used for Prediction	Evaluation of the Approach	Main Result
(Alhijawi et al., 2025)	AI Catcher using MLP and CNN	LLM generated scientific content detection	Custom ChatGPT generated dataset	Linguistic and statistical features	Accuracy and F1 score	Fusing linguistic patterns with deep features effectively detects ChatGPT scientific text
(Saqaabi et al., 2025)	Deep Learning for paragraph generation	Paragraph level paraphrasing detection	Custom paragraph dataset	Inter sentence and intra sentence relations	Accuracy and BLEU score	Identifies complex manipulations like sentence splitting and shifting missed by others
(Alnaimat et al., 2025)	Review of AI tool capabilities	Ethics checks in journal submissions	Not applicable	AI detection tool features	Not applicable for review	Journals require AI tools to automate ethics checks for incoming submissions
(Amirzhanov et al. 2025)	Systematic Literature Review	Taxonomy of plagiarism types	Scientific literature	Detection algorithms and plagiarism categories	Not applicable for survey	Defines modern plagiarism types including intelligent and cross lingual variants
(Antonius et al., 2023)	E BERT with Smith Waterman algorithm	Obfuscated plagiarism detection	PAN PC 11 dataset	Contextual semantic embeddings	Accuracy of 99.5 percent	Hybrid model achieves near perfect accuracy by aligning semantics with sequence
(Asalkar et al. 2025)	Hybrid BERT LSTM model	Scientific sentence similarity detection	Custom dataset	Word embeddings like TF IDF and OpenAI	Accuracy comparison	Hybrid BERT LSTM models outperform CNN based models for scientific text

Reference	Used Approach	Problem Domain Studied	Dataset Used	Attributes Used for Prediction	Evaluation of the Approach	Main Result
(Bao et al. 2024)	Attentive Siamese LSTM network	Low resource plagiarism detection	Tibetan Chinese dataset	Attention weights on word embeddings	Accuracy and F measure	Attention mechanisms improve detection by focusing on salient keywords
(Bouaine et al. 2023)	Doc2Vec with Siamese LSTM	Cross language plagiarism detection	PAN 11 and Europarl datasets	Contextual word embeddings	F1 score	Doc2Vec with LSTM outperformed BERT for English Spanish document pairs
(Chang et al. 2023)	Joint Coarse- and Fine-grained mechanism	Document similarity recognition	PAN PC 11 dataset	TF IDF and semantic vectors	F1 score	Two stage filtering improves processing speed without sacrificing detection accuracy
(El-Rashidy et al. 2024)	SVM with Chi square feature selection	Stylistic text plagiarism detection	PAN 2013 and PAN 2014	42 stylistic features like POS tags	Accuracy of 96.5 percent	Feature engineering with SVM effectively detects stylistic anomalies in text
(Greiner-Petter et al. 2025)	Evaluation of detection systems	Generative AI plagiarism in science	PAN 2025 Llama and Mistral data	Embedding vectors and similarity scores	Recall and Precision	Naive semantic similarity has high recall but low precision on older data
(Ilyasa et al. 2025)	Bibliometric Analysis using VOSviewer	Trends in AI plagiarism research	250 scientific articles	Keywords and citation networks	Not applicable for analysis	Research on AI plagiarism in education has grown significantly since 2015
(Islam et al. 2023)	Cosine Similarity and Jaccard	Low resource Bengali plagiarism	Custom Bengali dataset	TF IDF and word tokens	Accuracy and Precision	Established a baseline for detecting plagiarism in the low resource Bengali language
(Jing & Liu 2023)	DistilBERT with Differential Evolution	Population based plagiarism detection	MSRP and PAN PC 11	Distilled word embeddings	Accuracy and Training Time	Retained 97 percent of BERT accuracy while being 60 percent faster to train
(Lieharyani et al. 2025)	Sentence BERT vs Levenshtein	Homogenous student answer datasets	854 student answers	Semantic vectors vs lexical edit distance	F1 score and Recall	Lexical methods outperformed SBERT in homogenous short answer contexts
(Liu et al. 2023)	Scenario based SimCSE and SBERT	Text similarity optimization	Custom essay dataset	Contextual embeddings and N grams	Accuracy and Efficiency	Selecting models based on text complexity balances detection speed and accuracy

Reference	Used Approach	Problem Domain Studied	Dataset Used	Attributes Used for Prediction	Evaluation of the Approach	Main Result
(Moravvej et al. 2023)	BERT with Attention LSTM and DE	Imbalanced plagiarism classification	PAN 2013 and PAN 2014	BERT embeddings and attention weights	F1 score	Differential Evolution optimization improves performance on imbalanced plagiarism data
(Murch et al. 2025)	Turnitin AI detection analysis	AI prevalence in doctoral dissertations	200 dissertations from 2013 and 2023	Turnitin AI probability scores	Prevalence percentage	Found 6 percent of recent dissertations contained significant AI generated text
(Romanov et al. 2024)	Integrated NLP technique	Authorship verification in academia	Natural and artificial academic texts	Stylistic and linguistic features	Classification Accuracy	Method distinguishes between natural and artificial text based on writing style
(Sajid et al. 2025)	Comparative Review	Text based plagiarism techniques	Literature including English Novels corpus	Computational cost and accuracy	Not applicable for review	Traditional methods are faster but Deep Learning models offer higher accuracy
(Setu et al. 2025)	BERT and Cosine Similarity strategy	Plagiarism in academic submissions	Custom academic submissions	Semantic embeddings and TF IDF	Similarity threshold accuracy	Two step process of semantic check then direct copy check is effective
(Uhryn et al. 2025)	Machine Learning with Sentiment Analysis	Authorship identification	Textual content	Sentiment analysis features	Identification Accuracy	Sentiment features help identify true authors of textual content accurately
(Zahid et al. 2023)	Supervised SVM and Naïve Bayes	Cross lingual Urdu English plagiarism	Custom Urdu English dataset	Translated N grams and features	Classification Accuracy	Classifiers effectively detect translated plagiarism between Urdu and English
(Zhou et al. 2024)	Review of Deep Learning methods	Paraphrase identification	MSRP and PAWS datasets	Paraphrase type distributions	Not applicable for review	Standard datasets lack diverse paraphrase types limiting model generalization

Table 3 provides a systematic comparison of the 24 analyzed scientific articles, categorizing each study based on its algorithmic approach, problem domain, and experimental results. This comparison highlights the rapid methodological advancement in the field, specifically the shift toward hybrid deep learning architectures in 2025. Among the most promising recent developments, (Alhijawi et al., 2025) demonstrated how effective a hybrid MLP-CNN model ("AI-Catcher") can be, which successfully blends linguistic patterns with deep features to detect LLM-generated scientific content with high accuracy. (Saqaabi et al., 2025) presented a novel

deep learning approach that is capable of identifying complex paragraph-level paraphrasing such as sentence splitting and shifting which addresses a structural gap that sentence-level detectors often miss. Finally, (Asalkar et al., 2025) validated that hybrid BERT-LSTM models generally beat out traditional CNNs when it comes to capturing semantic similarity. This confirms the potential of combining Transformer embeddings with sequential learning for optimal detection performance.

2.4 Summary of Related Literature Work Analysis

In the comparative analysis the reated scientific articles in Table 3, it becomes obvious that there is a major methodological shift happening in the field. We are moving quickly from old-school feature engineering toward much more advanced deep learning architectures. For example, traditional methods like Support Vector Machines (SVM) are still quite good at spotting stylistic anomalies, often hitting accuracies near 96.5% (El-Rashidy et al., 2024). However, they really struggle when they run into the complex, semantic tricks used in modern scientific plagiarism. To solve this, (Bao et al., 2024; Bouaine et al., 2023) found that recent research has started leaning more on sequential models. LSTM networks, especially the ones using attention mechanisms, have proven to be much better at capturing those long-range dependencies that are vital for catching paraphrasing or even cross-lingual fraud. As of now, the absolute state of the art is defined by Transformer-based models. Hybrid approaches that mix BERT with sequence alignment algorithms to reach almost perfect accuracy around 99.5%, even when dealing with heavily obfuscated text (Antonius et al., 2023.). But there is a catch. Despite their high accuracy, these Transformer models come with significant computational costs. It promps a new wave of research into optimization techniques, such as DistilBERT and Differential Evolution that aim to balance the detection precision with processing efficiency (Jing & Liu, 2023.; Moravvej et al., 2023). Furthermore, the rise of AI-generated content has introduced an entirely new dimension to the problem by adding the necessity of novel strategies that do more than simple similarity checking to verify if the scientific discourse is actually authentic (Alhijawi et al., 2025; Murch et al., 2025).

3. CONCLUSION

This section presents the general conclusions as follows:

1. Looking back at the current state of text similarity detection, one thing is very clear. Traditional lexical and feature-based methods might be fast and efficient, but they simply are not enough to catch the complex tricks found in modern scientific writing. The review of the field highlights a major shift toward Transformer-based models. Specifically, BERT and its various optimized versions now represent the absolute state of the art. It is safe to conclude that these specific models are far more capable than the older tools we used to rely on. They excel at grasping bidirectional context and can spot those tiny, subtle variations in meaning that define intelligent plagiarism.

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